https://github.com/Et1enneee/comp3500-logistics

COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application serves one warehouse using Next.js, TypeScript, Prisma and SQLite. Real GPS, barcode scanning, recipient telephone collection, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5). Requirements and models link to evidence; §10 collects the trail.

2. Development process

Plan-driven increments have triggers and verified outcomes; integration and continuous integration provide feedback. Increments cover product and specification changes and recorded process milestones.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded
Test-led review, 17 Oct 2026	Assess order and stock functions	Defects recorded; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Failed endpoint and missing ledger	Transaction verified
Integrated application, 1 Nov 2026	Integrate foundation	Customer, staff and admin workflows
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Assigned tasks and data-minimisation checks
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned
Export simplification, 11 Nov 2026	Scripted walkthrough finding (issue #15)	Selected-date export in three actions; regression passed

The workflow, issue tracker and increment register record integration.

3. Requirements

The specification defines obligations and rationale.

Functional requirements and verification

FR	The system shall	Rationale	Verification
FR-01	create orders with product lines and totals	start fulfilment	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	protect invariants	Test — lifecycle and stock
FR-03	number orders and shipments and show owner tracking	retrieve progress	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	support replenishment	Test — inventory and alerts
FR-05	record event notifications and allow reading	communicate progress	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	support review	Test — aggregates and export
FR-07	authenticate and enforce role and object access	protect data	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	complete handoff	Test — assignment and delivery

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 integration availability (green build–test–browser runs recorded per increment); NFR-03 least privilege (role gates on every mutating endpoint); NFR-04 ≤3 actions per scripted operation; NFR-05 process quality (CI-verified increments).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	Increment register	scheduling plan
RK-02 contract rework	FR-02, NFR-05	transition tests
RK-03 test-week capacity	Priority register	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	FR-01/04/06, NFR-05	seed regression test
RK-06 assignments and test squeeze	Risk plan	assessment schedule
RK-07 stack unfamiliarity	NFR-05 support	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

Five views link roles, lifecycle, data, transaction, layers. Local transactions preserve consistency without splitting services; state and stock guards enforce rules. The four layers — Role-specific pages, HTTP routes and authentication, Order/inventory/driver business handlers, Prisma/SQLite persistence — each isolate one concern.

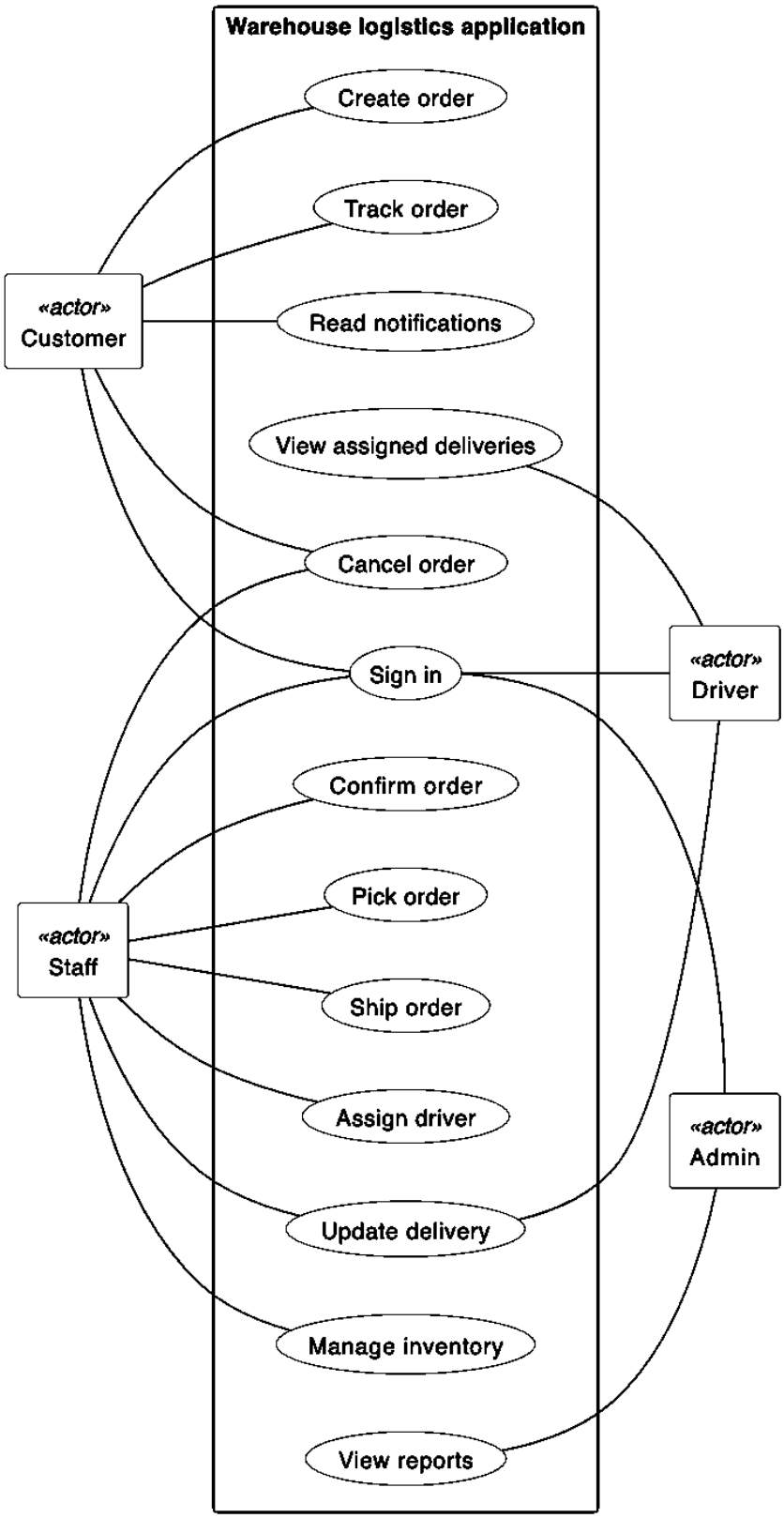


Figure 1. Which roles perform each function? Source.

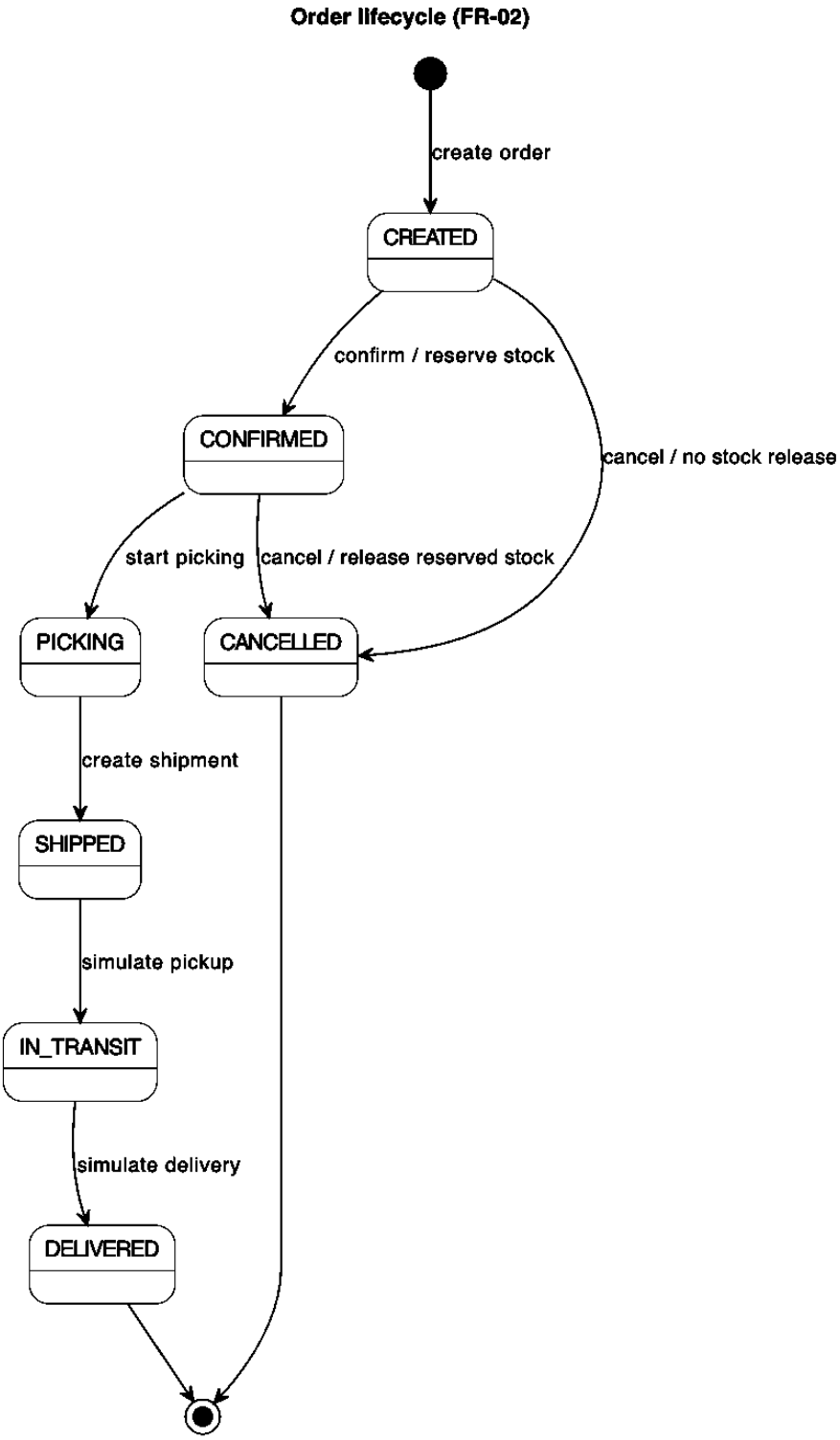


Figure 2. Which order transitions and stock effects are valid? Source.

Each item references one product; each order has 0..1 shipment; each product has 0..* inventory logs.

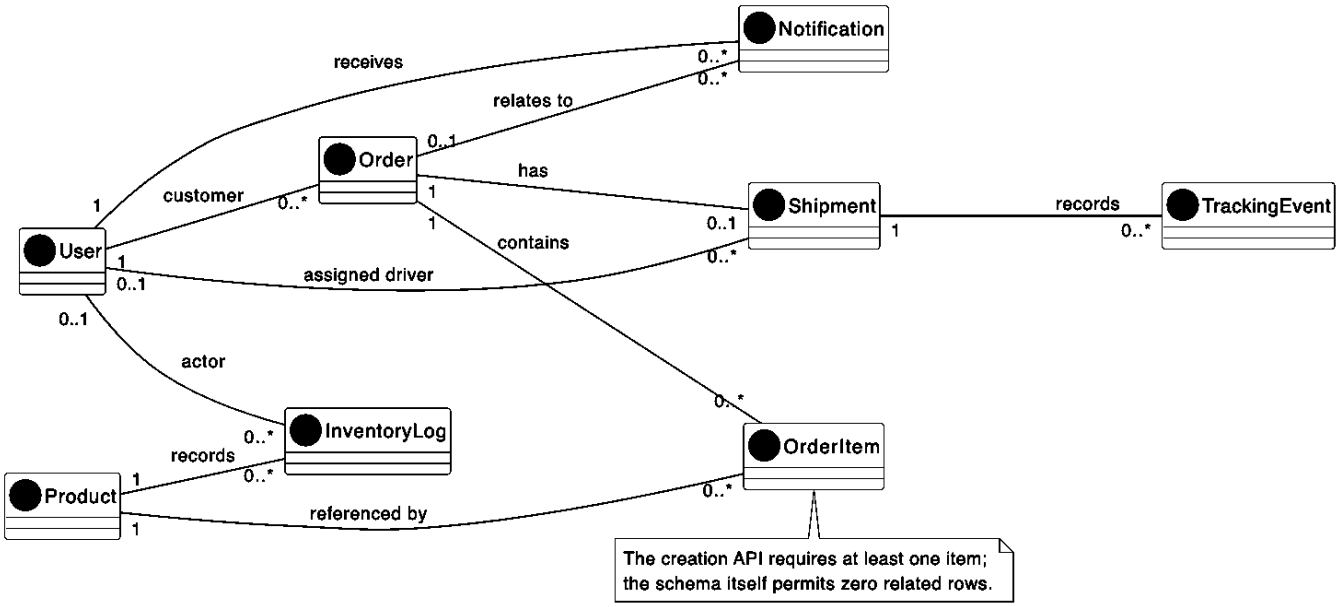


Figure 3. Data relationships: each order requires at least one item, enforced at the creation API. Source.

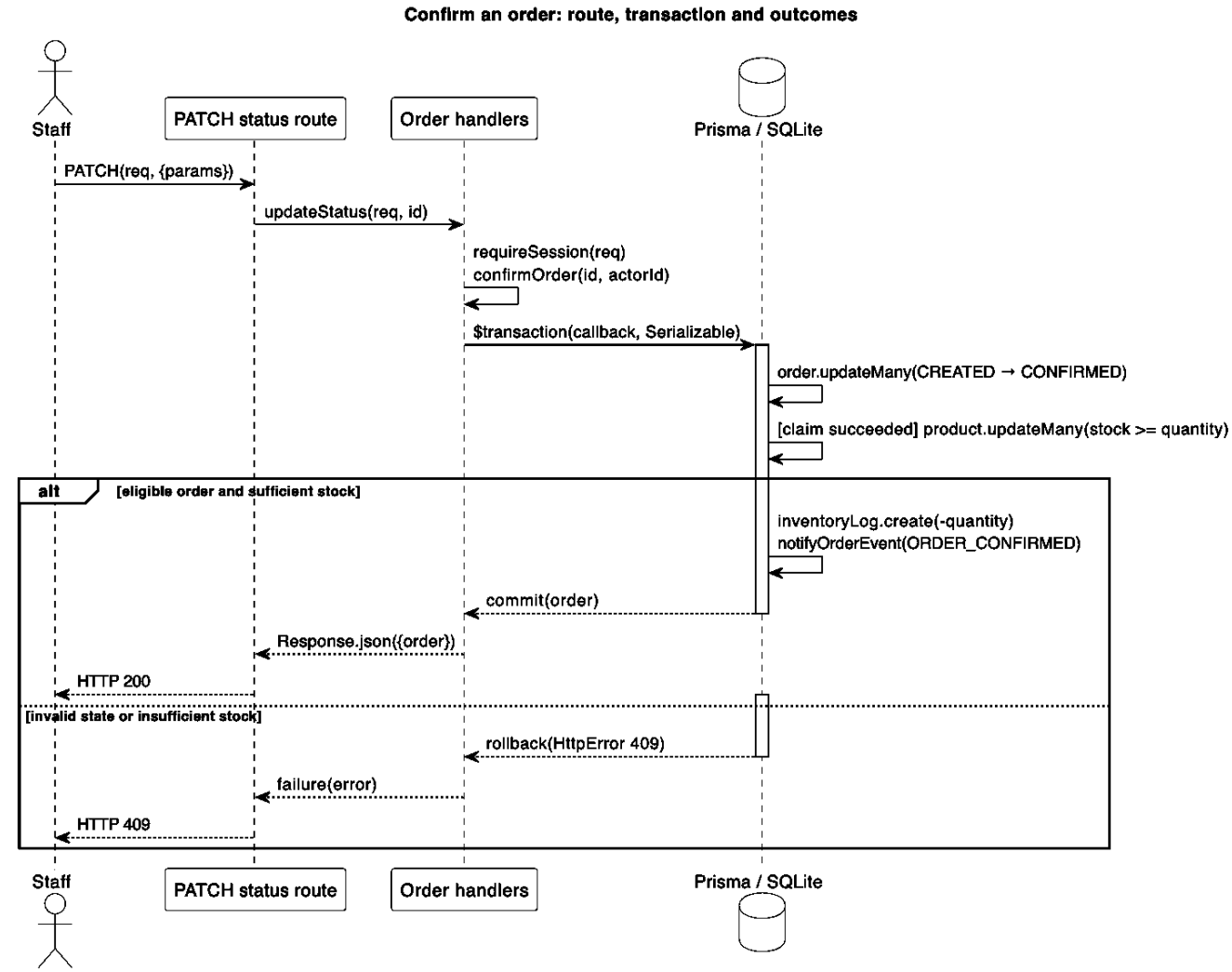


Figure 4. How does confirmation commit or roll back? Source.

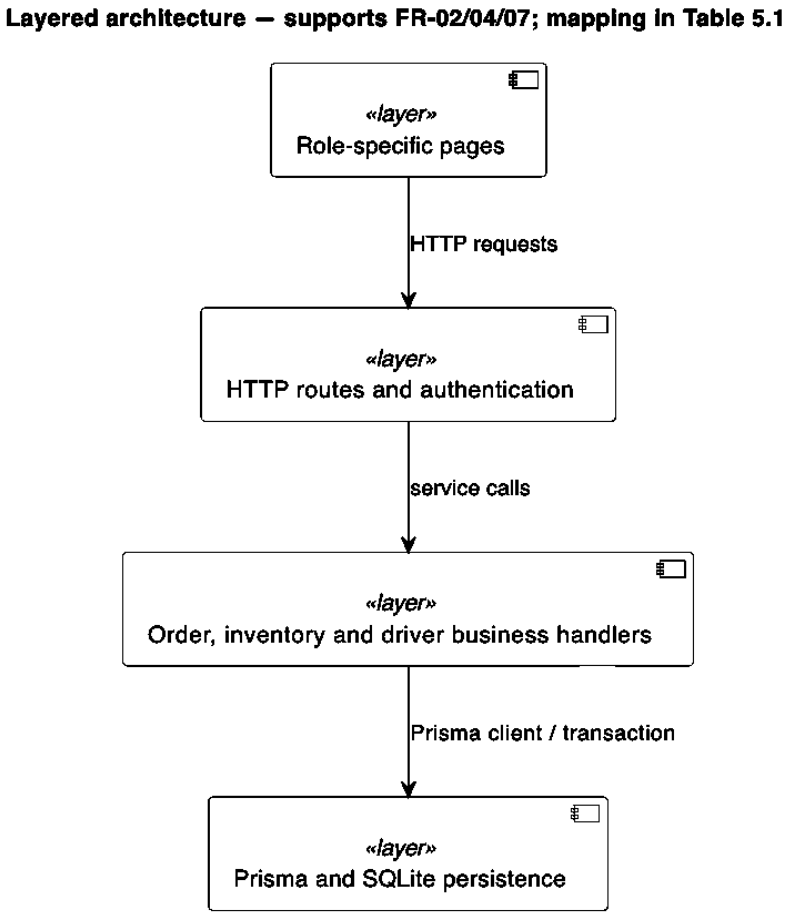


Figure 5. How do four layers connect? Source.

5. Implementation

Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED, then IN_TRANSIT and DELIVERED; CANCELLED from CREATED (no restock) or CONFIRMED (restock), rejected after PICKING. Customers cancel their own; staff cancel CONFIRMED.

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Serializable state, stock, ledger and notification writes	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
Selected-date export	admin/page.tsx:24–48	Browser

6. Verification and validation

Verification checks requirements against the implementation; validation exercises the delivered workflows through the interface. The regression log and CI show 38 unit, 140 integration, 2 delivery (driver simulation) and 26 browser checks.

Boundary case	Observed	Test
Repeat confirmation	HTTP 409; state and stock unchanged	Orders
Stock equals demand	Confirmation succeeds; stock becomes zero	Orders
Stock 4 versus threshold 5	Alert; stock 5 is not low	Inventory

Insufficient stock returns HTTP 409 with unchanged state and stock. In a local-loopback measurement recorded on 11 November 2026, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. NFR-02 is evidenced by the recorded green build–test–browser runs (observations).

7. Evolution

The early defect record drove confirmation and retest; the documentation hardening review corrected models and renewed tests. An administrator export task became three actions (regression).

8. Validation

The automated walkthrough exercises eleven core operations through the interface as each of the four roles, at most three actions each, with expected outcomes asserted in the walkthrough script; results are logged with the run.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Evidence
Diagrams	Corrected models and architecture	Compliance matrix
FR-04	Stock transition effects belong to FR-02	Traceability
Increments	Triggers and outcomes	Register
Admin task	Export re-tested; scripted walkthrough records the task (§8)	Issue #15
Unit count	Two availability-summary regression tests added, bringing the suite to 38	Test history
NFR scope	Performance and availability criteria set for local single-warehouse deployment	SRS revision log

10. Evidence

Traceability · Specification · Evidence directory · Baseline record.

Full evidence trail: docs/final-report.md; baseline `6eb01afb1798c7361205b10597d80c0c26e8fdf6` (code, evidence and test logs).